Proton–Electron Cosmology

RealQM and Newtonian Gravitation from a Single Electric-Potential Seed

Claes Johnson[1]

[1]KTH Royal Institute of Technology, Stockholm, Sweden , claesjohnson@gmail.com

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Mathematical theory and formulation by the author; manuscript developed in collaboration with Claude (Anthropic). See Section 10.

Abstract

Coulomb is the small-scale initial seed; Newtonian gravitation is the large-scale emergent consequence. We sketch an exploratory cosmology built from the RealQM / RealNucleus program, in which all matter is Coulombic (charge clouds in ordinary Euclidean space, no strong force, no elementary neutron) and all gravitation is Newtonian. The novelty here is a **single primitive seed**: a small, small-scale fluctuation of the electric *potential* φ_E alone — charge being its Laplacian, $\rho_c = -\varepsilon_0 \nabla^2 \varphi_E$. From it the universe develops in two tiers. **At the small scale**, the Laplacian inflates the potential seed into charge of both signs and of *different spatial extent* — small dense positive (protons) and large diffuse negative (electrons); since a contracted cloud carries more energy, that extent *is* the inertial mass (E/c^2), and Coulomb geometry then builds nuclei (RealNucleus) and atoms (RealQM). **At the large scale**, the *energy* of that charged matter is a mass/inertia density ρ, and it is this ρ — seeded entirely by the charges — that sources gravitation through Poisson, $\rho = \nabla^2 \varphi_g$. Read about the mean, ρ carries both signs, so the gravitational source is two-fold; the perturbations separate into great chunks of positive mass *interfoliated* with negative mass — the cosmic web of filaments and clusters interleaved with under-dense voids, the negative sector acting as a repulsive dark energy. There is thus *no primitive negative mass and no separate gravitational seed*: only the electric potential is seeded, mass is the energy of the charged world, and gravity is emergent. The construction is strictly non-relativistic — Coulomb below, Newton above — so the speed of light c enters only the electromagnetic sector: there is a local speed limit for charges and light, but none for the large-scale gravitational motion of neutral bodies (two galaxies may approach at relative speed $> c$). We state the picture, its one quantitative handle (a single transition energy $E_* \approx 1.29$ MeV — the electron's compact–expanded gap — fixing the primordial helium fraction), and its load-bearing open problems — chiefly *what sets the nuclear scale.*

Status. An exploratory proposal, not established cosmology. It departs from the Standard Model and ΛCDM and must eventually be reconciled with them; the open problems of Section 9 are load-bearing. Read it as "what follows *if*."

Keywords: cosmology; two-tier cosmology; proton–electron model of the nucleus; emergent gravitation; dark energy; primordial nucleosynthesis; RealQM.

1 Introduction

The RealQM program formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by free boundaries, minimising the Coulomb energy [1]. Its nuclear extension, RealNucleus, revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interactions, with no strong-force postulate [2]. This note asks what cosmology results if the *same* ingredients are taken as the starting point of the universe, together with ordinary Newtonian gravitation.

The answer is a **two-tier** cosmology from **one seed**. The seed is a fluctuation of the electric *potential* (charge being its Laplacian). It shapes the small scale directly (charge $\to$ nuclei $\to$ atoms, all of RealQM); and the *energy* of that charged matter, being mass, sources an *emergent* Newtonian gravitation that shapes the large scale (the cosmic web, with a repulsive negative-mass sector as dark energy). We keep strictly to what Coulomb and Newton can say, mark the extreme regimes (strong gravity, the highest energies) where more would be needed, and keep the open problems explicit.

2 The machinery: two-valued Poisson, opposite force-signs

Both forces share *one* structure and differ in a single sign. Each is the gradient of a potential obeying Poisson's equation with a two-valued ($\pm$) source,

$$\text{electric:}\quad \rho_c = -\varepsilon_0\,\nabla^2\varphi_E, \qquad\qquad \text{gravitational:}\quad \rho_m = \frac{1}{4\pi G}\,\nabla^2\varphi_g, \tag{1}$$

the force being $-(\text{charge})\,\nabla\varphi$ in each case. In this cosmology the electric potential is the primitive seed (Section 3) and the gravitational source is the emergent energy of the charged world (Section 6) — but the *machinery* is common. The one structural difference is the relative sign, and with it the force law:

	like sources	unlike sources
electric (charge)	**repel**	attract
gravitational (mass)	**attract**	repel

Concretely there are two independent binary signs — the source $\pm\nabla^2\varphi$ (two-valued) and the force $\mathbf{F} = \pm\nabla\varphi$ (repulsive or attractive for like sources) — and their *four* combinations are exactly the four "charges" the two forces couple to: the repulsive-for-like sign is electromagnetism, its $\pm$ source the $\pm$charge of the small scale; the attractive-for-like sign is gravitation, its $\pm$ source the $\pm$ mass/energy of the large scale. One operator and two signs thus yield $\{\pm\text{charge}, \pm\text{mass}\}$.

This single sign difference generates the two tiers. Because like charges *repel* and unlike attract, matter neutralises into $\pm$ pairs, charge cancels, and electromagnetism *screens* — it stays local and small-scale. Because like masses *attract*, mass *clumps* and accumulates, growing without bound into large-scale structure; and because unlike masses repel, positive- and negative-mass regions *separate* into the interfoliated web and voids of Section 6. The scale separation is thus a *consequence* of the force-sign, not a separate assumption: small-scale electromagnetism and large-scale gravitation are one two-valued Poisson structure, read with two signs. (This is the static Poisson level; the electromagnetic sector carries c and radiation, and $\pm$-mass gravitational dynamics has its own subtleties — both belonging to the extreme regimes outside the Coulomb–Newton scope kept here.)

A second asymmetry — in the *direction* the Poisson relation is used — fixes the *scale* of each force. For electromagnetism the *potential* is primary: the seed is a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — and charge is obtained by *differentiating*, $\rho_c = -\varepsilon_0\nabla^2\varphi_E$. The Laplacian is a *local* operator that *magnifies* short wavelengths ($\rho_k \propto k^2\varphi_k$), so precisely because the seed is small-scale (large k) it is inflated, despite its tiny amplitude, into sharp, small-scale $\pm$ charges. For gravitation the *source* is primary — the mass/energy of the charged world — and the potential is obtained by *integrating*, $\varphi_g = 4\pi G\,\nabla^{-2}\rho_m$. The inverse Laplacian is a *global* operator (convolution with the $1/r$ Green's function) that *smooths*, suppressing short wavelengths ($\varphi_k \propto \rho_k/k^2$) and building a broad, large-scale potential. So the same Poisson relation, run forward makes small-scale electromagnetism and run backward makes large-scale gravitation: **the Laplacian acts locally and magnifies; the inverse Laplacian acts globally and smooths** — and that is why charge is small and gravity large.

3 The arena and the single seed

The arena is a plain **3D Euclidean coordinate space** — absolute space, simply given, with no physical medium. The only operator needed is the Laplacian ∇^2, and the only primitive is a small-scale fluctuation of the **electric potential** φ_E. Charge is not postulated separately; it is the curvature of that potential,

$$\rho_{\text{charge}} = -\varepsilon_0 \, \nabla^2 \varphi_E, \tag{2}$$

with two immediate consequences. **Both signs:** ρ_{charge} is positive where φ_E is concave, negative where convex. **Net zero:** $\int \nabla^2 \varphi_E \, dV$ is a surface term that vanishes, so the universe carries **zero net charge** automatically — equal numbers of protons and electrons, with no baryogenesis. And because the Laplacian multiplies each Fourier mode by k^2 ($\rho_k \propto k^2 \varphi_k$), a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — becomes a large density contrast: the seed is tiny in amplitude but short in wavelength, and it is the short wavelength (large k) that the Laplacian amplifies into strong charge.

Crucially, **only the electric potential is seeded** (charge being its Laplacian). Mass and gravitation are not primitive; they arise, in Section 6, as the *energy* of the charged structures the seed produces. This is the essential departure from a cosmology of two independent potentials: here there is one seed, one potential, and gravity is a consequence.

4 The small scale: RealQM and RealNucleus

The seed, inflated by the Laplacian, produces charge of both signs and, decisively, of *different spatial extent*: small dense positive charge and large diffuse negative charge. These are the **proton** and the **electron**, and the whole charged world is built from them by Coulomb geometry.

Size is inertial mass. In RealQM a particle is a charge cloud whose size is fixed by the balance of a localisation (kinetic) cost against Coulomb attraction. A *contracted* cloud costs more localisation and Coulomb energy and so carries *more energy*; and inertia is energy content, mass = E/c^2 [3]. Hence **small = heavy, large = light**: the proton (small, dense) is heavy, the electron (large, diffuse) is light. "Proton small, electron big" and "proton heavy, electron light" are one fact, not two — a size difference read through E/c^2. (The numerical ratio $m_p/m_e = 1836$ is not derived; it is the open input of Section 9, as it is in every theory.)

No overlap, no self-interaction: self-repulsion is kinetic energy. RealQM charge clouds do not overlap — each electron occupies its own domain with a free boundary — and carry *no self-interaction*: there is no Coulomb self-repulsion term. The role such self-repulsion would play, resisting the collapse of a charge to a point, is carried instead by the *localisation energy* $\frac{1}{2}\int |\nabla\psi|^2$: compressing a cloud sharpens its gradients and costs kinetic energy, so it resists compression and settles at a finite size (localisation cost against Coulomb attraction). A free electron, unconfined, therefore spreads; a bound one takes the size where localisation balances attraction. In RealQM, **self-repulsion appears as kinetic energy**.

Nuclei and atoms, one mechanism at two scales. With a neutron taken as $p+e$, a neutral atom (A, Z) holds A protons and A electrons: $A - Z$ *compact* inside the nucleus and Z *expanded* in orbitals. The electron occupies two size-phases, its extent adapting to the Coulomb well: compact in a deep well (inside a nucleus, $\sim$fm, $\sim$MeV) and expanded in a shallow one (atomic,

~Å, ~eV). Chemistry and nuclear physics are then one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by expanded (valence) electrons; a nucleus is protons glued by compact electrons. RealNucleus reproduces nuclear binding this way — the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7, from Coulomb geometry alone [2], and light-nucleus fusion (hydrogen burning) appears as a rearrangement of protons and electrons, not the action of a separate force.

The nuclear/atomic scale. What sets the two sizes can be read from charge and the proton's finite size R_p alone, with *no mass and no c*. An electron attracted to one proton in vacuum must taper to zero at infinity; the localisation cost then forces a large cloud of size the atomic $a_0 = \hbar^2/me^2 \approx 0.5\,\text{Å}$. An electron caged among protons at a multiphase free boundary need *not* taper to zero — its density hands off to the proton density — so a flat caged density has zero localisation cost, and it sits at the cage (proton) size $R_p \approx$ a few fm, bound by Coulomb (~MeV at fm). The size ratio is therefore

$$\frac{a_0}{R_p} \;\approx\; \frac{1}{\alpha^2} \;\approx\; 2\times 10^4, \tag{3}$$

so the fine-structure constant appears here as a pure *geometric* ratio of the two scales, $\alpha^2 = R_p/a_0$, the ratio of nuclear to atomic size — not a coupling constant. Empirically the H atom ($\approx a_0 = 0.53\,\text{Å}$) and the deuteron nucleus ($\approx 2.1\,\text{fm}$) give a ratio $\approx 1/25{,}000$, matching $\alpha^2 \approx 1/18{,}800$ to order of magnitude. *Why* the proton is ~fm — what fixes R_p — remains the central open problem.

5 Inertia is energy; the local speed limit

Throughout, inertia is the energy content: $\mathbf{p} = (E/c^2)\mathbf{v}$, the measured relation of the Penning trap [3], and Newton's second law $\mathbf{F} = \dot{\mathbf{p}}$ governs the response. In the small, the force is electromagnetic and couples to *charge*; in the large, it is gravity and couples to *mass* $= E/c^2$. One mechanics, one inertia, two forces reading two properties of the same matter.

The constant c enters *only* the electromagnetic sector — through the field momentum of a moving charge, Thomson's electromagnetic mass. Consequently there is a **local speed limit**: a charge, or a pulse of light, cannot overtake light *through* the local matter and fields, because feeding it energy only increases its inertia E/c^2 without bound as $v \to c$. But **Newtonian gravitation is c-free** — instantaneous, with no c in $\rho = \nabla^2\varphi_g$ — so it imposes *no* speed limit on the large-scale motion of neutral bodies. Two galaxies may approach, or recede, at relative speed *greater than c*; this is a global relative velocity of gravitating matter, not a local signal, and it violates no local causality (no local frame sees anything pass light through it). The speed limit is thus a property of the RealQM/Maxwell sector, not of gravity — just as in observational cosmology distant galaxies recede faster than light without any local law being broken. (How Newtonian mechanics extends to relatively moving frames — where c and its kinematics would enter — is a separate matter, taken up in *Many-Minds Relativity* [4], and lies outside the non-extreme scope kept here.)

6 The large scale: emergent Newtonian gravitation

The energy of the charged matter — protons, electrons, their binding and radiation — is a mass/inertia density ρ, **positive throughout** ($\rho \geq 0$), seeded entirely by the charges. It is this positive ρ that sources gravitation, and the emergent potential φ_g comes in **two forms**, its Laplacian taking either sign against the one positive source:

$$\pm\frac{1}{4\pi G}\,\nabla^2\varphi_g \;=\; \rho \;\geq\; 0. \tag{4}$$

Thus φ_g is not a primitive but *emergent*, the large-scale field of the aggregated energy of the charged world, and it comes in two forms **according to the sign of its Laplacian**: the + form is a *well* ($\nabla^2\varphi_g > 0$, effective positive mass, attractive), the − form an *anti-well* ($\nabla^2\varphi_g < 0$, effective negative mass, repulsive). The ± mass is therefore *formal* — the sign carried by the curvature of φ_g, not a negative energy; the source ρ is positive. By the gravitational force-sign, the wells attract and draw together while the anti-wells repel, so the positive-mass regions are **isolated** into great chunks — filaments and clusters — *interfoliated* with the anti-well regions between them: the **cosmic web**, positive-mass nodes and filaments interleaved with voids, the anti-well (formally negative-mass) sector acting as **dark energy**.

Both the positive and the negative mass are emergent — the two forms of the gravitational potential of the one charge-seeded, positive energy ρ — neither a primitive, neither a second seed; they repel and **separate**, the positive isolating into the web and the negative emptying into the voids. Only charge is put in; the whole mass–gravity structure, of both signs, is the emergent large-scale field of that energy. This is the sense in which the largest structures and the smallest share one origin: the small scale is the seed and its charge directly, the large scale the emergent gravity of their energy.

A 2D test: cosmic-web morphology. A mass-conserving 2D simulation of a ±mass field $\rho = \nabla^2\varphi$ from a random potential, evolved as compressible self-gravitating flow with same-sign attraction and opposite-sign repulsion, develops the morphology of observed large-scale structure (Figure 1): intermixed positive- and negative-mass filamentary webs, mass concentrated at nodes, separated by large voids — the voids here *actively driven* by negative-mass repulsion rather than merely evacuated by gravity. A qualitative match (a 2D toy); quantitative tests (void-size distribution, correlation function) remain to be done.

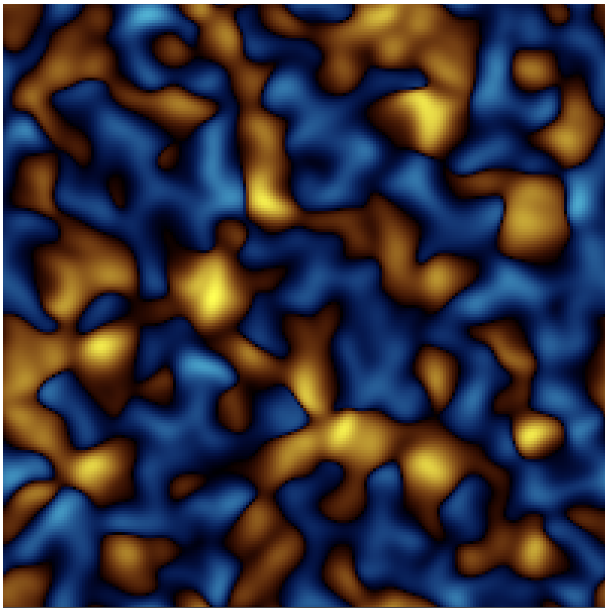

Figure 1: 2D self-gravitating ±mass from $\rho = \nabla^2\varphi$: positive- (warm) and negative- (cool) mass webs with concentrated nodes and large voids — cosmic-web morphology, voids driven by negative-mass repulsion. Mass-conserving; periodic box.

7 The cosmic sequence and the helium fraction

The two sectors exchange energy in *both* directions, and the cosmic sequence is that exchange in time. *Upward*, the energy of the charged world is mass and sources gravity (Section 6). *Downward*, gravitational attraction feeds energy back into electromagnetism: the collapse of a positive-mass region converts gravitational potential energy into heat, and that hot dense state drives the electromagnetic processes. Gravity supplies the energy; electromagnetism spends it as structure and light.

Concretely: ordinary gravitational contraction of a positive-mass seed supplies one hot dense epoch, and the released binding energy leaves as light. As the contracting proton–electron matter heats, **deuterons and nuclei** form ($\sim$MeV, RealNucleus), the Coulomb binding energy radiating away (the origin of light) and appearing as the mass defect. On cooling, nuclei capture expanded electrons into **atoms** ($\sim$eV, RealQM), again radiating; at that recombination the matter turns neutral, electromagnetism screens out, and radiation streams free. Only *then* does the weak, slow gravitational aggregation build the cosmic web of Section 6. So the universe's timeline stages the two tiers in order: *charge builds the small while matter is charged; gravity builds the large once it is neutral.*

A bound state survives once the bath can no longer ionise it, so the two electron phases switch on at two temperatures set by their two binding energies, forcing the order plasma → nuclei → atoms (Table 1). The temperature ratio between the epochs ($\sim 10^6$) is just the energy-scale ratio.

epoch	process	temperature
plasma	free ± charges	$> 10^{10}$ K
nuclei	compact e captured → nuclei	$\sim 10^{10}$ K ($kT \sim$MeV)
atoms	expanded e captured → atoms; radiation	$\sim$3000 K ($kT \sim$eV)

Table 1: Two condensation epochs, separated by $\sim 10^6$ in temperature.

The quantitative handle: the helium fraction. The model carries a single free number, and it fixes the primordial helium. The electron has two size-phases — *compact* (inside a nucleus, i.e. inside a neutron, since neutron $= p +$ compact e) and *expanded* (atomic) — interconverted by the two weak processes, compactify ($p + e \to n$) and expand ($n \to p + e$, β-decay). Three quantities enter:

- E_* — the **energy gap** between the two phases: the energy it costs to compactify an electron, released again when it expands. It plays the role the neutron–proton mass difference plays in standard cosmology and numerically equals it, $E_* \approx 1.29$ MeV. This is the model's *single free handle.*
- T_* — the **transition temperature**, $T_* \sim E_*/k \sim 10^{10}$ K: above it the two phases freely interconvert in thermal equilibrium, below it compactification becomes rare.
- T_f — the **freeze-out temperature** (≈ 0.8 MeV): the point where the size-changing (weak) processes fall slower than the cosmic expansion and can no longer maintain equilibrium, so the compact/free — i.e. neutron/proton — ratio *locks.*

While in equilibrium the neutron/proton ratio is the Boltzmann factor $e^{-E_*/kT}$; at freeze-out it locks at $(n/p)_f = e^{-E_*/kT_f}$. A fraction of the free neutrons then β-decay back before nucleosynthesis (a survival factor d), and essentially all the rest end bound in He-4:

$$\left(\frac{n}{p}\right)_{\text{nuc}} = \frac{(n/p)_f\, d}{1 + (n/p)_f(1-d)}, \qquad Y = \frac{2\,(n/p)_{\text{nuc}}}{1 + (n/p)_{\text{nuc}}}. \tag{5}$$

With $E_* \approx 1.29$ MeV, $T_f \approx 0.8$ MeV (so $E_*/kT_f \approx 1.6$) and $d \approx 0.74$: $(n/p)_f = 0.199 \to (n/p)_{\text{nuc}} = 0.140 \to Y = 0.245$ — the observed primordial helium, the model's first falsifiable quantitative success. (As in standard big-bang nucleosynthesis T_f is not itself predicted but set by the balance of weak rate against expansion — a second input — so the genuine prediction is the *ratio* $E_*/kT_f \approx 1.6$, with E_* the single MeV-scale handle.)

8 Complexity from the size asymmetry

The world is built not on a charge imbalance (charge is exactly balanced, $\pm e$) but on a *size* imbalance: the proton has a fixed small size (a rigid anchor) while the electron's size is variable, adapting to its well — ∼Å in an atom, ∼fm in a nuclear cage. The fixed proton supplies the anchors; the variable electron *spans scales* — spreading to bind atoms and molecules, compacting to bind nuclei — producing the nested multi-scale hierarchy that *is* complexity. With a single fixed size only one scale exists and nothing complex is built. The nuclear/atomic separation is thus a **precondition for complexity**, so any universe that supports structure, and hence observers, must exhibit it: an anthropic reason *why* the scales are separated, though not a derivation of *how far*. Notably, since a neutron is structurally $p + e$, it is heavier than a proton *by construction*, so proton and hydrogen stability is automatic — the usual fine-tuning bound "α must not be too large or the proton decays" simply does not arise.

9 Open problems

1. **The nuclear scale.** What sets the $\sim 10^4$–10^5 nuclear/atomic separation — equivalently, the proton size R_p? The preferred reading takes R_p (∼fm) as the input nuclear length, with the caged, zero-localisation electron sitting at it (no relativity needed); the ratio $a_0/R_p \approx 1/\alpha^2$ then follows. Why R_p has that value is open. This is the same datum as the mass ratio m_p/m_e and the proton/electron size ratio: one open number.

2. **The fluctuation spectrum.** $\rho = \nabla^2\varphi$ gives a blue (k^2-amplified) density spectrum, whereas large-scale structure fits a near scale-invariant one; the potential must carry a compensating red spectrum, and φ needs its own law of motion (here $\rho = \nabla^2\varphi$ is a definition, not yet a dynamics).

3. **The mass ratio $m_p/m_e = 1836$.** Built in qualitatively as a size difference, but its value is not derived (nor is it in the Standard Model).

4. **The neutrino.** β-decay's continuous spectrum — the 1930 objection that sank the proton–electron neutron [5]. The "electron expands" step needs a home for that energy and spin.

5. **Precision and scope.** BBN abundances beyond helium, the fluctuation-to-structure statistics, and the extreme regimes (strong-field gravity, the highest energies) lie beyond the Coulomb+Newton scope kept here.

10 Collaboration

The mathematical theory and the cosmological proposal are the author's. The manuscript was developed in dialogue with Claude (Anthropic), which also assisted in articulating the argument, surfacing the quantitative constraints and open problems, and preparing the supporting interactive simulations of the RealQM/RealNucleus program. We present the human–AI collaboration transparently as part of the working method.

One-sentence summary. On a given 3D Euclidean space, if the sole primitive is a small-scale fluctuation of charge — yielding small heavy protons and large light electrons, hence nuclei and atoms by Coulomb geometry (RealQM/RealNucleus) — then the *energy* of that charged matter, single quantity, sources an emergent Newtonian gravitation whose $\pm$ perturbations separate into the cosmic web (positive mass) and dark energy (negative mass), with the speed of light limiting only the electromagnetic sector; a unifying two-tier picture from one seed, whose survival hinges on a single hard question: what sets the nuclear scale.

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